ISAIAS MARTINEZ

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EDUCATION

Stanford University School of Engineering
Bioengineering with Honors, B.S. — Sep 2022 – Jun 2026
Computer Science, M.S. (Artificial Intelligence) — Sep 2024 – Jun 2027

EXPERIENCE

IBM Verify — Software Developer Intern — **Jun 2026 – Sep 2026**
- Built agentic AI systems for enterprise identity and access management, combining LLMs with automated reasoning to automate identity risk analysis
- Built a multi-agent identity risk workflow in Langflow routing cases across 4 specialized agents behind Python services and APIs, cutting end-to-end analysis from ~20 minutes of manual review to under 2 minutes per case
- Designed an MCP-based tool-filtering layer that scopes an agent's available actions to a user's entitlements, cutting exposed tool surface by over 15% across 5 workflows; traced ±8% run-to-run behavior variance to tool-ordering nondeterminism and reduced it to ±3%

Inspirit AI — AI Instructor — **Jun 2026 – Present**
- Teach Python, machine learning, and AI concepts to high school students through project-based instruction, leading small groups of 5-6 students per cohort
- Mentor students in designing and implementing end-to-end AI projects applied to fields such as education and investment using real-world datasets, building practical, portfolio-ready skills
- Provide technical guidance on model development, evaluation, and optimization

Medtronic Diabetes — Research & Development Intern — **Jun 2025 – Aug 2025**
- Automated test data collection and visualization for a next-generation glucose analyzer using Python and REST APIs, cutting manual data-processing time by 45%
- Rebuilt the test-data ETL path with secure pipelines and Git-based versioning, cutting runtime from ~25 minutes to ~6 minutes per batch; benchmarked device performance with statsmodels and scikit-learn across 200+ test runs

Stanford Medicine Medical Kiosk — Software Developer — **Jan 2025 – Present**
- Build multilingual intake modules on REST APIs and SQL backends, covering 4 languages across 3 intake flows
- Refined intake UX through feedback loops and A/B testing, cutting mid-intake abandonment from 22% to 9%
- Integrate medical-device APIs behind Dockerized microservices for scalable deployment and cross-device synchronization

Stanford Computer Science Department — Teaching Assistant — **Sep 2023 – Jun 2026**
- Mentored 10+ students per quarter across 6+ quarters in C++, Python, and JavaScript, teaching code design and debugging practices
- Debugged and reviewed 100+ code submissions quarterly, catching recurring design issues and promoting version control and testing methodology

RESEARCH

Stanford Marsden Cardiovascular Biomechanics Lab — Honors Thesis — **Feb 2025 – Jun 2026**
- Built a patient-specific pipeline from 4D flow MRI to CFD as sole student researcher: reconstructed 15 Fontan anatomies, screened to 3, generated paired virtual-stent geometries in svMorph, and ran 12 transient simulations at 32,000 time steps each
- Tuned branch-specific RCR Windkessel outlet conditions until simulated flow splits and mean pressures matched catheterization and MRI data; simulated moderate exercise (3× IVC inflow, 120 bpm) alongside rest
- Virtual stenting cut pressure drop 34.4–97.9% and resistance 70.4–99.5% across all three patients, but power loss rose 45.6% in one under exercise, showing a wider pathway is not automatically a more efficient one

Gordian Knot Center for National Security Innovation — Project Founder & Lead — **Sep 2025 – Jun 2026**
- Fused 3 independent data sources into a single risk index, validated against 50+ synthetic scenarios spanning baseline, outbreak, and stress-test conditions; selected an interpretable linear fusion model over higher-capacity alternatives to preserve explainability for decision-makers
- Defined precision, recall, and latency thresholds; achieved 88% precision / 82% recall at 150ms median latency

Stanford Fallacy Detection CS Research Project — Researcher — **Jun 2023 – Dec 2023**
- Benchmarked LLMs for fallacy detection; built baselines in Python (transformers, scikit-learn) with precision/recall/F1 reporting
- Curated datasets and prompts; implemented ETL, feature engineering, and reproducible eval pipelines; co-developed a fallacy detector and performed ablation studies and error analysis to refine model design

PROJECTS

Parameter-Efficient Adaptation of GPT-2 — *PyTorch, Transformers*
Built a GPT-2 Transformer from scratch and benchmarked LoRA, ReFT, and soft prompts against full fine-tuning — ReFT matched or beat full fine-tuning accuracy using 0.02% of trainable parameters.

Biosurveillance "Digital Immune System" — *Python, Streamlit, Pandas*
Built an early-warning dashboard fusing wastewater viral load, hospital admissions, and air-quality signals into a single biosecurity risk index with tiered alerts.

CoQuest — Spontaneous Social Coordination — *React Native, TypeScript, Firebase*
Built a full-stack app for students to broadcast and join spontaneous hangouts in real time; 10 users generated 67 quests in a 3-day pilot.

Brain Injury Connection — Accessible Redesign — *Product Management, UX*
Led an accessible redesign shipped across 8 pages/flows as PM with Develop for Good, incorporating feedback from 10+ user-testing participants.

SKILLS

Languages: Python, C++, JavaScript/TypeScript, SQL, Go
ML/AI: PyTorch, HuggingFace, LoRA/PEFT, sparse autoencoders, LLM evaluation harnesses, MCP
Frameworks/Libraries: React Native, NumPy, SciPy, Matplotlib, Streamlit, Firebase, Node.js
Tools: scikit-learn, statsmodels, pandas, Git/GitHub, Docker, CI/CD, Linux, REST APIs, SQL, PostgreSQL, SimVascular, Langflow
Systems/Infra: Microservices, distributed systems, cloud APIs (AWS/GCP), Bash

AWARDS

Stanford Award of Excellence
Stanford Gordian Knot Center National Security Innovation Scholar
HSF Scholar
SHPE GE Scholarship